

## **IIOT for Advanced Manufacturing (ECMC104)**

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### **I. Smart Greenhouse Monitoring System**

### **II. Smart Museum Preservation System**

#### **Intended Learning Outcomes**

1. Build applications on edge devices to enable sensing and control.
2. Configure equipment for network connectivity.
3. Build applications on edge devices to enable IoT connectivity.
4. Create dashboard for data visualisation.
5. Build an end-to-end IoT application.

#### **Weightage**

This project contributes 50% of the coursework.

#### **Deadline**

Week 13 (Interim Report Submission)

Week 14 (Interim Assessment)

Week 17 (Final Assessment and Report Submission)

#### **Contents**

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|          | AI Assessment Scale Level Permitted:  | 5        |

# 1 Introduction

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## I. Smart Greenhouse Monitoring System

A Smart Greenhouse Monitoring System is becoming increasingly important in modern agriculture and smart farming industries. A greenhouse management company plans to improve its greenhouse operations by implementing a smart greenhouse monitoring system. The company has engaged your project team to provide an IoT-based solution for the greenhouse facility.

One of the main functions is to maintain stable and consistent temperature and humidity levels within the greenhouse environment. This is important to support healthy plant growth and maintain suitable growing conditions.

The company is looking for the following features in the system:

- Greenhouse staff can monitor the temperature and humidity levels of the greenhouse through smart devices at all times.
- Greenhouse staff can receive alerts when the temperature or humidity exceeds predefined threshold values.
- Greenhouse staff can remotely control cooling or ventilation devices using smart devices.

Example:

Greenhouse staff can remotely switch on a cooling or ventilation fan (simulated by the 5V fan) when the temperature or humidity becomes too high.



## II. Smart Museum Preservation System

A Smart Museum Preservation System is becoming increasingly important in museums and heritage conservation industries. A museum management company plans to improve the preservation of valuable artefacts and exhibits by implementing a smart museum preservation monitoring system. The company has engaged your project team to provide an IoT-based solution for the museum facility.

One of the main functions is to maintain stable and consistent temperature and humidity levels within the museum environment. This is important to protect artefacts, paintings, and historical collections from environmental damage and deterioration.

The company is looking for the following features in the system:

- Museum curators can monitor the temperature and humidity levels of exhibition and storage areas through smart devices at all times.
- Museum curators can receive alerts when the temperature or humidity exceeds predefined threshold values.
- Museum curators can remotely control cooling, heating, or ventilation devices using smart devices.

Example:

Museum curators can remotely switch on a cooling or ventilation system (simulated be the 5V fan) when the temperature or humidity becomes too high.



## 2 Task

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To develop a Smart Greenhouse / Museum solution.

Develop the prototype with the following hardware and software components complete with a IoT data visualisation platform which will eventually lead to a complete IoT system.

- BME280 Temperature, Humidity and Pressure Sensor
- Buzzer, 5V fan
- Raspberry Pi as edge device and gateway
- MQTT for data and message transport
- Code in Python

Additional components or add-on features should be added to improve the overall process automation. In addition to data visualisation, you should also consider the analytics layer.

A well-designed user interface with an easy-to-use dashboard is an important aspect of the IoT system.

## 3 Deliverables

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The project deliverables include the following:

- To demonstrate a working prototype.
- To write a report on the project.

Each member will need to explain his/her contributions during the interview.

The report should contain the following sections:

### 1. Introduction

- Describe the problem based on the given problem statement.
- Describe the proposed solution.

### 2. Project Plan

- Formulate a plan for completion of the project, e.g. identify the sub tasks and set a timeline for each sub task.

### 3. IoT System Design

- Design the system architectural diagram for the smart system.
- Design the I/O pin assignment diagram for the Raspberry Pi I/O interface.
- Provide details of the MQTT protocol (publisher, subscriber, topic, data transported)
- Design a schema for the time-series database.
- Provide details of the dashboard for monitoring of the data.

### 4. Problems Encountered/Recommendation

## 5. Individual Reflection

- Write your reflection on the learning experience for the project. You may include the problems encountered and how you resolve those issues in the write-up.

## 6. Source codes with comments and clear explanation

## 7. References

# 4 AI Assessment Scale

### AI Assessment Scale Level Permitted:

|          |                             |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3</b> | <b>AI<br/>COLLABORATION</b> | <p>AI may be used to help complete the task, including idea generation, drafting, feedback, and refinement. Students should critically evaluate and modify the AI-suggested outputs, demonstrating their understanding.</p> <p><b>You may use AI to assist with specific tasks such as drafting text, refining and evaluating your work. You must critically evaluate and modify any AI-generated content you use.</b></p> |
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### Guidelines for ethical use of AI:

- Use AI responsibly to support your learning
- Acknowledge any GenAI used
- Show your own understanding and ideas
- Verify AI-generated information for accuracy
- Be able to explain your work